

Optimization and reliability. Exam 04th June 2019

Exercise 1: 13%

A bakery has 150 kg of flour, 22 kg of sugar and 15 kg of butter. He has two popular recipes for making a dozen of cakes:

A cakes: 1 dozen A-cakes can be made of 3kg of flour, 1 kg of sugar and 1.25 kg of butter

B cakes: 1 dozen B-cakes cab be made of 5 kg of flour, 0.5 kg of sugar and 0.5 kg of butter

The profit of a dozen A cakes is 20 kr, while it is 30 kr for a dozen of B-cakes.

- Formulate the optimization problem of maximizing profit using these two recipes (for a dozen cakes) according to the available amounts of flour, sugar and butter. Use the standard form.
- Find the solution to the problem using the graphical method (make simple sketch)

Exercise 2: 12%

We will consider gradient-based minimization of the following unconstrained function:

$$f(x) = 2x_1 + (2 + x_1)^2 + (1 - x_2)^2$$

The starting point is: $x(0)=[1,2]^T$

- Complete the first iteration of the steepest descent method for the function.
The 1D line search problem should be solved analytically
- Complete the first iteration of the modified Newton's method for the function.
The 1D line search problem should be solved analytically.

Exercise 3: 8%

Solve the following linear optimization problem using the basic steps of the SIMPLEX method and tableau's:

Minimize:

$$f(x) = -x_1 - 10x_2$$

Subject to constraints:

$$g_1(x) = -x_1 + x_2 \leq 4$$

$$g_2(x) = 4x_1 + x_2 \leq 20$$

$$x_i \geq 0 \quad \forall x_i = \{1,2\}$$

Exercise 4: 8%

The following optimization problem is considered:

Minimize: $f(x) = 6x_1^2 + 4x_2^2 - 10x_1x_2 - x_2 + 2$

Subject to constraints: $g_1(x) = x_1 + x_2 \leq 0$

- Use the alternate KKT necessary conditions to determine the candidate minimum points
- Use the sufficient conditions to determine the minimum point(s). It is informed that a non-zero feasible direction for the optimum is $d=[1 \ 1]^T$

Exercise 5: 9%

Solve the following problem by setting up the solution tree and using the Local Minimization Brach & Bound method:

Minimize: $f(x) = -5x_1 - 4x_2$

Subject to: $g_1(x) = -x_1 + x_2 - 5 \leq 0$

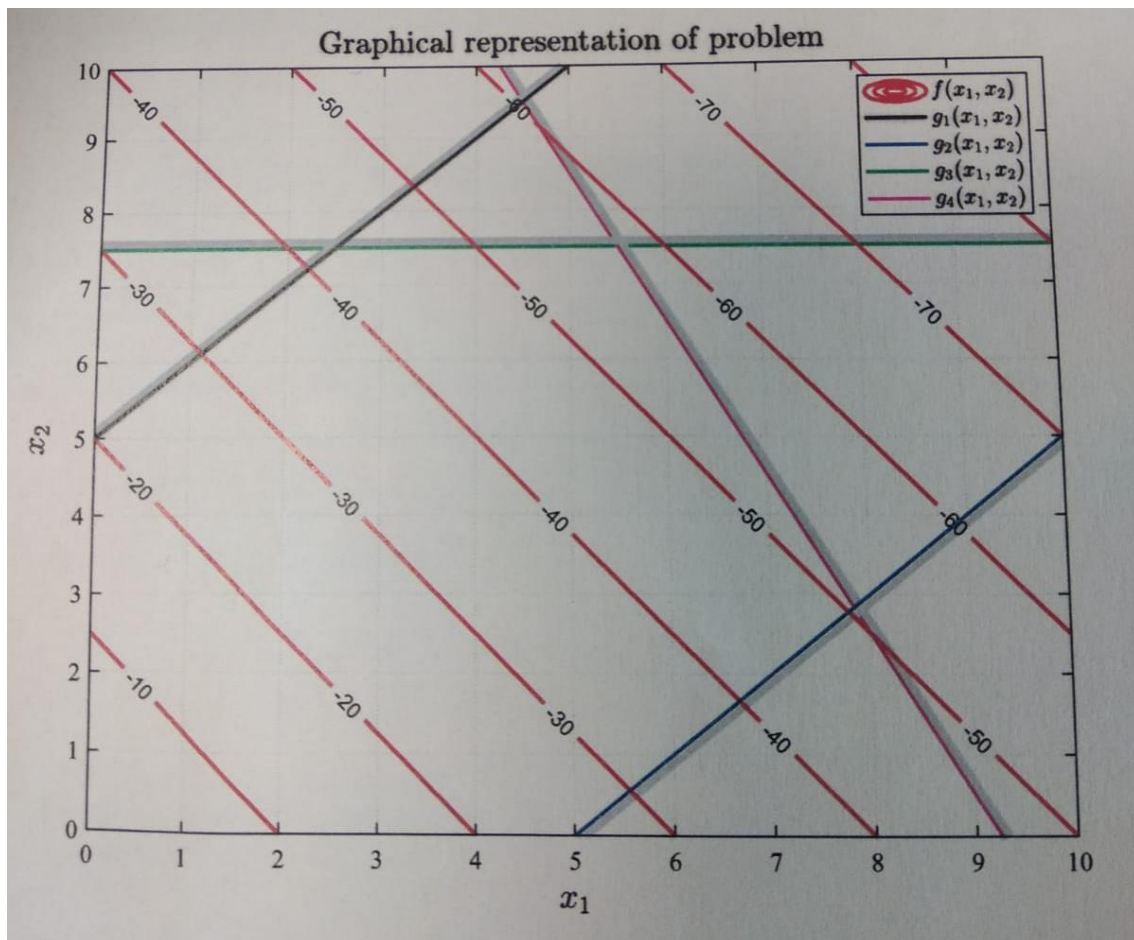
$$g_2(x) = x_1 - x_2 - 5 \leq 0$$

$$g_3(x) = x_2 - 7.5 \leq 0$$

$$g_4(x) = 2x_1 + x_2 - 18.5 \leq 0$$

Both x_1 and x_2 should be integer values

As help for solving the problem the objective function contours and constraints are plotted in Fig 1.



Exercise 6: 14%

There are two kinds of rechargeable batteries in the market from suppliers A and B, with the same voltage level and capacity. In order to compare the reliability performance of these two types, the field operation experiences of 6 samples for each type were collected. All the 12 samples are assumed to have same usage conditions in terms of environmental conditions, frequency of charging and charging schemes. The recorded time-to-failure data are shown in the table below:

Sample no.	1	2	3	4	5	6
Type A (days)	1050	1250	1400	1500	2000	2100
Type B (days)	1300	2150	3250	4000	5150	6900

- 1) (5%) Plot the time-to-failure curves of Type A and Type B batteries in the specific usage conditions in the attached Weibull paper on next page. Please find out the values of the Weibull parameters β and η for the Type A and Type B batteries, respectively.
- 2) (5%) Based on the Weibull plots, please find out:
 - a. (2%) What are the B1 lifetime of Type A and Type B batteries with 50% confidence level?
 - b. (2%) What are the B10 lifetime of Type A and Type B batteries with 50% confidence level?
 - c. (2%) What are the average lifetime of Type A and Type B batteries with 50% confidence level?
- 3) (4%) If you are asked to select one of the two types of batteries for a reliability-critical application that should limit the total percentage of failure from battery to 1% with 50% confidence level, which type will you choose and why?

Appendix I – Median rank table
sample size = n
failure rank = i
n

i	1	2	3	4	5	6	7	8	9	10
1	.5000	.2929	.2063	.1591	.1294	.1091	.0943	.0830	.0741	.0670
2		.7071	.5000	.3864	.3147	.2655	.2295	.2021	.1806	.1632
3			.7937	.6136	.5000	.4218	.3648	.3213	.2871	.2594
4				.8409	.6853	.5782	.5000	.4404	.3935	.3557
5					.8706	.7345	.6352	.5596	.5000	.4519
6						.8906	.7705	.6787	.6065	.5481
7							.9057	.7979	.7129	.6443
8								.9170	.8194	.7406
9									.9259	.8368
10										.9330

Exercise 7: (11%)

A wooden pencil consists of four sub-systems of wooden shaft, lead shaft, eraser and ferrule. In order to prepare the information used for Failure Mode and Effect Analysis (FMEA), please answer the following questions. The answers of the questions will be evaluated mainly on the understanding of the procedures and NOT on the accuracy of the wording of specific descriptions.

- 1) (3%) Assume the pencil is the system of interest. Please plot its boundary diagram to show at least two interfaces of it.

- 2) (4%) Plot the Parameter diagram (P-diagram) of pencil to include the main aspects of a P-diagram (i.e. input signals, ideal response, noise factors, control factors, and error states).
- 3) (4%) Identify two failure modes and two failure causes of the pencil and one of the four subsystems, respectively.

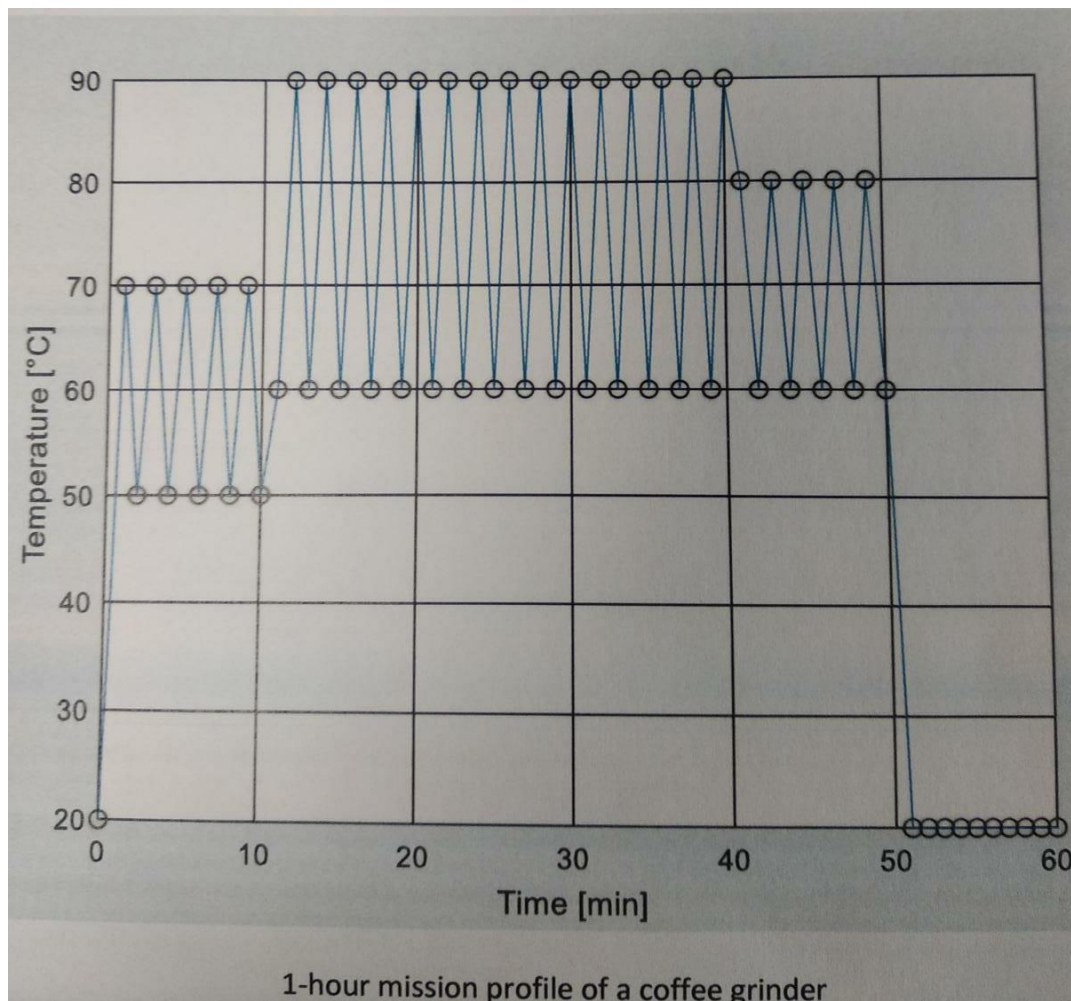
Exercise 8: (12%)

A jet engine comprises two main parts: a compressor and a turbine. For a correct operation, both parts have to work. The compressor manufacturer provides a Weibull reliability function with $\beta_1=2.6$ and $\eta_1=175000\text{h}$. The turbine manufacturer provides an exponentially distributed reliability function with $\text{MTBF}_2=127000\text{h}$.

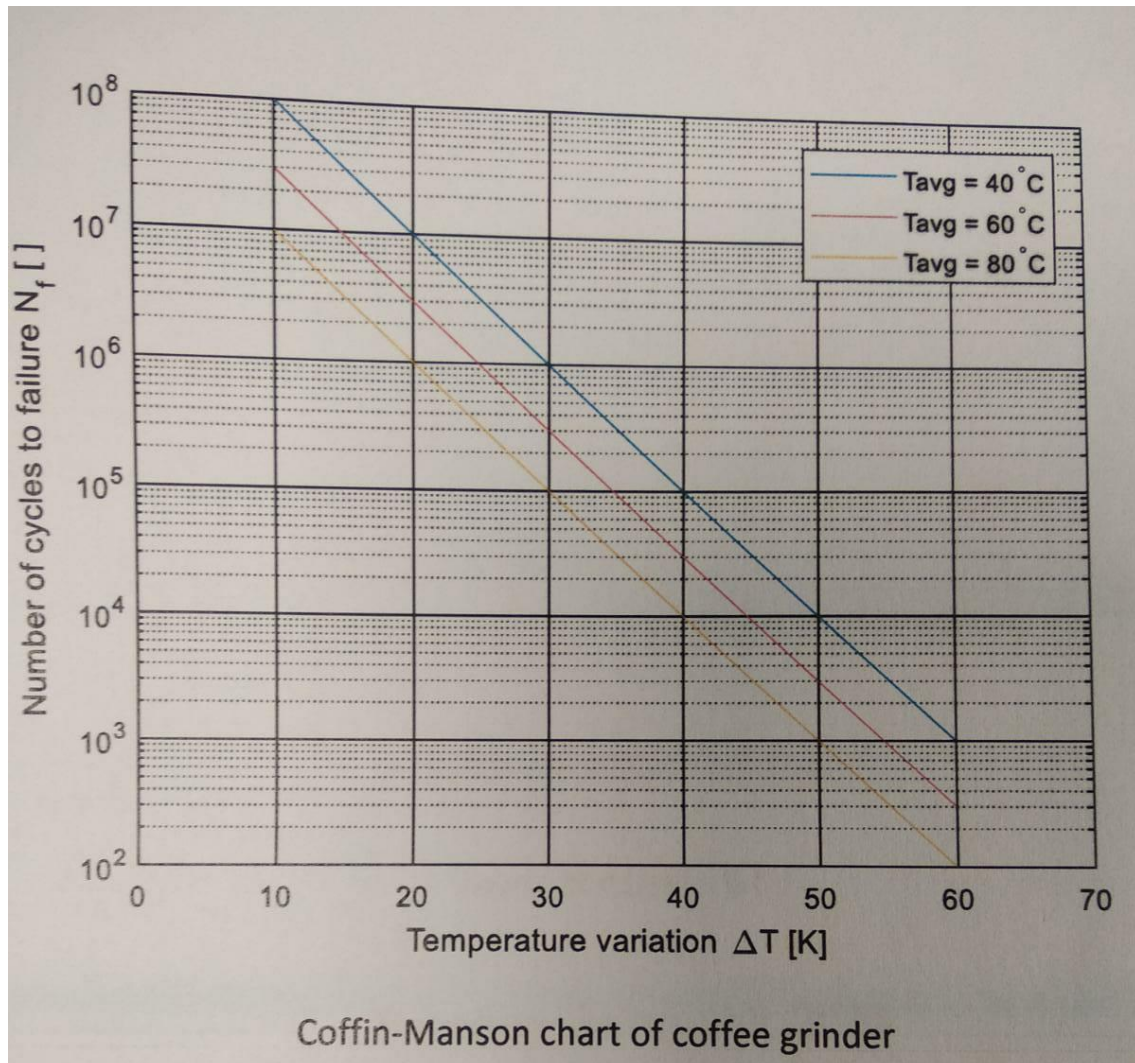
- a) (2%) Sketch up an RBD schematic of the overall system
- b) (5%) Calculate the reliability of each component after 1 year
- c) (3%) Work out the overall reliability function, and the overall system reliability after 1 year
- d) (2%) Calculate or give a good estimation of the overall system β_1 lifetime.

Exercise 9: (13%)

The 1-hour mission profile of a coffee grinder is reported in the following figure, where the temperature is plotted against time. The same profile is repeated 6 hours per day, 5 days per week.



The coffee grinder lifetime has been evaluated and modelled as Coffin-Manson lifetime model, i.e. taking into account temperature average (T_{avg}) and variation (ΔT), as shown in the following chart.



- 1) (2%) Identify and list in a table the three different stress levels (T_{avg} , ΔT) occurring in the given position profile
- 2) (3%) Identify such levels as points in the Coffin-Manson chart and complete the above table with the number of cycles to failure N_f for each of them
- 3) (6%) Calculate the accumulated damage (in %) within 1 hour
- 4) (2%) Calculate the expected lifetime in years